

Timothée Herbeau

Master's student in Fundamental Physics

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Education

2024 – 2025

Master's degree in Fundamental Physics

Université Paris Saclay – Physics Dept.

Relevant classes: Advanced Statistical and Quantum Physics, Particle Physics, Non-Linear Physics, Phase Transitions, Condensed Matter & General Relativity.

2022 - Present day

Master's degree in Engineering

École des Mines de Paris - PSL University

Relevant classes: Statistical & Quantum Physics, Continuum mechanics & Special Relativity. Optimisation, ODE, Computer Science & ML. Advanced calculus, advanced linear algebra.

2020 - 2022

Preparatory classes - Classes préparatoires

Lycée Saint-Louis - Paris

PC*-PCSI

Intense science (Mathematics, Physics, Chemistry) and humanities classes leading to national exams (written and oral) to enter engineering schools.

Experience

2025 (3 months)

LPTMC - Jussieu



Investigate the theoretical model of a 2D anisotropic harmonic oscillator with anisotropic non-gaussian noise. Tutored by Pascal Viot.

2024 (5 months)

Safran



Fluid mechanics modeling of hydraulic pumps used in the latest development of primary flight controls. Use of continuum mechanics to assess sources of inefficiency and potential resonance phenomena.

2023-2024 (5 months)

MIT - Mechanical Eng. Dpt. – Boston



Modelled a flame-assisted spray-pyrolysis device to optimize its working conditions. Confronted the simulations to experiments. Developed the new generation of that device based on the simulations. Weekly presentations to the lab.

In parallel, established a mathematical model and coded a MATLAB simulation of the chemico-physical process of regeneration in a diesel particle filter.

2023 (3 months)

Mines de Paris & Framatome



Coded a Python tool to predict temperature profiles in nuclear power plant ducts to be used by Framatome. Team of four students.

2021-2022 (1.5 years)

Lycée Saint Louis – Paris



Conducted a study on phononic crystals, with numerical simulations and experiments. Produced a thesis and defended it.

Computer science

Experienced

C++

Python : NumPy, Pandas Sklearn

Office

Ansys Fluent

Intermediate

Git

LaTeX

SolidWorks

MATLAB

Julia

Beginner

Flask

HTML/CSS/JS

Languages

French

(mother tongue)

English

(C2)

Spanish

(B2)

Chinese

(A1)

Hobbies

Personal projects

French Physicist Tournament: member of University Paris Saclay's team for the 2025th edition (final rank: 2nd)

Member of the English debating team of University Paris Saclay.

Member of the committee of review of the physics curriculum at Mines de Paris

Tutoring of high school students in mathematics and physics

Member of the environmental student association at Mines de Paris (NOISE)

Sports

Kickboxing (7 ans)

Volleyball (5 ans)

Music

Guitar (6 ans)